Manuscript outline

Christophe Rouleau-Desrochers

June 9, 2026

Introduction

1. The most frequently observed biological impact of climate change is major changes in phenology (Parmesan & Yohe, 2003; Cleland *et al.*, 2007; Lieth *et al.*, 1974; Woolway *et al.*, 2021; Menzel *et al.*, 2006)
   (a) Shifts in spring and autumn phenology modifies when seasons start and end (Duputié *et al.*, 2015).
   (b) Understanding the consequences of these shifts requires understanding how much and why growing seasons have changed (Duputié *et al.*, 2015).
   (c) Spring phenology advances, (Wolfe *et al.*, 2005; Chmielewski & Rötzer, 2001; Fu *et al.*, 2014) driven by temperature (Chuine, 2010; Cleland *et al.*, 2007; Peñuelas & Filella, 2001)
   (d) Autumn phenology delays, but much less than spring (Gallinat *et al.*, 2015; Jeong & Medvigy, 2014) and driven by temperature and photoperiod (Cooke *et al.*, 2012; Flynn & Wolkovich, 2018; Körner & Basler, 2010; Delpierre *et al.*, 2016)
2. Shifts in spring and autumn phenology supports the assumption that longer growing seasons increase tree growth (Keenan *et al.*, 2014; Stridbeck *et al.*, 2022)
   (a) Recent research shows that this may not be the case (Dow *et al.*, 2022; Green & Keenan, 2022; Silvestro *et al.*, 2023; Wang *et al.*, 2025; Matula *et al.*, 2023; Wang *et al.*, 2025)
   (b) Dow *et al.* (2022) showed that trees did not grow more or faster despite warmer and earlier springs at the regional scale
   (c) Cabon *et al.* (2022) showed a decoupling between GPP and carbon storage at the global scale.
   (d) If trees don't grow as we expect, it could affect forest carbon cycle projections (Richardson *et al.*, 2013; Swidrak *et al.*, 2013) and subsequent climate solutions (Green & Keenan, 2022)
3. Understanding these finds requires answering why trees do not grow more despite longer growing seasons.
   (a) We have a poor understanding of carbon allocation to above-ground biomass, despite it being one of the largest carbon sink (Cabon *et al.*, 2022).
   (b) Linear representation of growth as a function of carbon assimilation is wrong (Cabon *et al.*, 2022)
   (c) Trees may be source or sink limited, which emphasizes the importance of understanding the temperature sensitivity relationship between growth and photosynthesis (Gessler & Zweifel, 2024).
   (d) Growth may be more sensitive to varying environmental conditions than photosynthesis, which could mislead models (Cabon *et al.*, 2022, 2020; Muller *et al.*, 2011; Peters *et al.*, 2021)
   (e) While warmer spring temperatures advance the onset of cell production, that alone does not increase wood production with water playing a more decisive role on wood prod (Wang *et al.*, 2025).
   (f) Complex nature of climate change makes understanding the internal physiological and external constraints to growth critical (Almagro *et al.*, 2025; Cabon *et al.*, 2022; Wolkovich *et al.*, 2025)
4. How trees internal programming and their susceptibility to environmental conditions could constrain their growth?
   (a) Growth as any other biological mechanism is genetically controlled, and will thus vary by species (Howe *et al.*, 2003; Wolkovich *et al.*, 2025; Silvestro *et al.*, 2025)

(b) Tree plasticity is a highly variable trait that changes with species' life history strategies (Cuny *et al.*, 2019; Michelot *et al.*, 2012), affecting how flexible their growth is in response to varying environmental conditions.

(c) While CO_2 concentrations increased, drought, heat waves, competition, insect outbreaks and spring frosts—factors projected to change in frequency and severity with a warming climate—can offset the benefits of longer seasons (Tyree & Zimmermann, 2002; Choat *et al.*, 2018; Li *et al.*, 2023; Trenberth *et al.*, 2014; Change, 2014; Chiang *et al.*, 2021; Polgar & Primack, 2011; Reinmann *et al.*, 2023).

(d) Droughts can act as a severe growth constraint (Kang *et al.*, 2023) by stopping growth early (Dox *et al.*, 2022) or inducing tissue mortality (Li *et al.*, 2023).

(e) Intra- and inter-specific competition for nutrients, water and light can also offset a tree's capacity to take advantage of extra days in the season (Cleland & Wolkovich, 2024).

5. Understanding the constraints to growth requires defining the growing season and growth.

   (a) Different definitions of GS: we will use calendar (number of days) and thermal season (window of opportunity of thermally favorable meteorological conditions for plant growth) (Körner *et al.*, 2023) (Figure 1)

   (b) Calendar season only accounts for the number of days in the season, implying a stable growth rate throughout the season.

   (c) This may reveal a role of some neglected cooler and shorter days in the season that have a weak influence on the thermal season (Figure 1a).

   (d) Growth rate changes with temperature, which the thermal season accounts for and should therefore better predict growth responses.

   (e) Wood formation (xylogenesis) is the allocation of carbon for long-term storage (Etzold *et al.*, 2022; Silvestro *et al.*, 2025), and the radial growth increments are represented through tree rings (Plomion *et al.*, 2001)

   (f) Allometries using tree DBH and height have a coarse temporal scale and can miss extreme events affecting growth (e.g., Meyer, 1940; Saunders & Wagner, 2008).

   (g) Tree rings are more precise growth proxies, which we use here (Gazol *et al.*, 2018; Manzanedo & Pederson, 2019)

6. Growth drivers differences across species need to be considered.

   (a) Phenology and the relationship between growth and growing season length varies heavily across species (Etzold *et al.*, 2022)

   (b) Pooling all species together in carbon sequestrations models is a weakness (Green & Keenan, 2022; Cabon *et al.*, 2022; Wolkovich *et al.*, 2025)

   (c) We propose to use ground-based leaf phenology observations across several species to address this issue (Piao *et al.*, 2019)

   (d) These observations provide valuable insights of the growth temporality of trees not suitable for experimental trials (Wolkovich *et al.*, 2025)

   (e) Cambial phenology in diffuse-porous trees is closely synchronized with leaf phenology (Marchand *et al.*, 2021; Silvestro *et al.*, 2025; Suzuki *et al.*, 1996), so we can use them to as a reliable proxy for start and end of the growing season (Stridbeck *et al.*, 2022)

7. Local adaptation could inform growth responses of trees under future climate change (Soolanayakanahally *et al.*, 2013; Wolkovich *et al.*, 2025).

   (a) Provenance refers to the geographic and climatic origin of a population (Aitken & Bemmels, 2016; Housset *et al.*, 2018)

   (b) Trees from northern provenances sometimes have later start of season (Zeng & Wolkovich, 2024) and often earlier end of season phenology = shorter growing season (Way & Montgomery, 2015; Vitasse *et al.*, 2014).

(c) This is especially true for budset—marks the end of primary growth—, with its phenotypic variance being strongly related to provenance and photoperiod acting as the primary cue (Aitken & Bemmels, 2016; Soolanayakanahally *et al.*, 2013; Way & Montgomery, 2015).

(d) Thus, common gardens with provenances across a latitudinal gradient can inform tree growth responses to future climatic conditions.

8. To answer if trees grow more with longer growing seasons, we related three years of leaf phenology and growth data across four species and four provenances in a common garden on juvenile trees. Our objectives are:

   (a) Determine if trees grow more with longer thermal vs calendar season (ref: conceptual figure).

   (b) Understand the calendar growing season relationship with growth and how it relates to the SOS and EOS.

   (c) Determine how growth changes across years, provenances and individual trees.

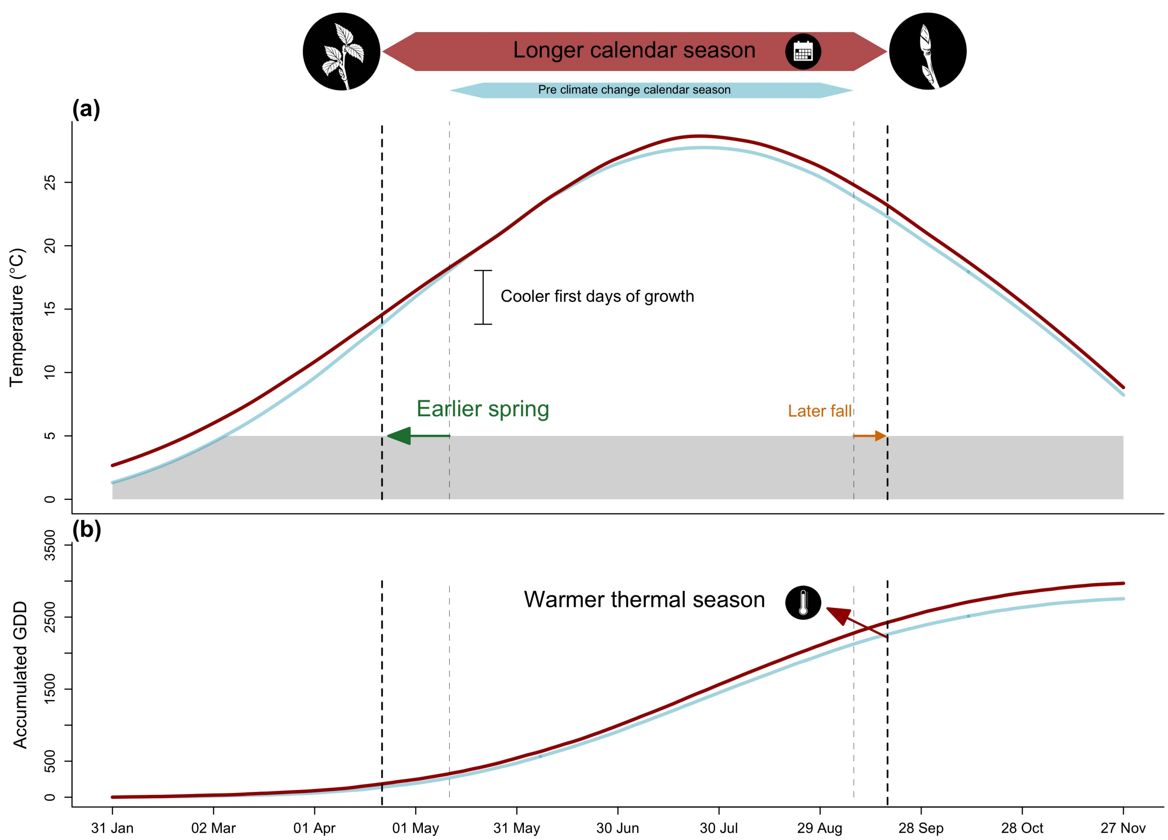

Figure 1: We show how shifting growing season length with climate change affects both the temperature trees experience during the growing season (a), and how much growing degree days (GDD) they accumulate (b). We used mean daily temperature data from the Boston airport Climate station for the smoothed temperature and GDD curves. We represent the changes in temperature before (1945-1965 in blue) and after the 1980s increase in climate warming (2005-2025 in red). With climate change, spring events have advanced and fall phenology has delayed, but to much lesser extent (represented through the arrows), this lengthened the calendar growing season, but concurently changed the temperature trees experience early in their season. As a results of increased temperature during the growing season, trees also accumulate more GDD, warming their thermal seasons.

Results

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
   (b) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
   (c) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
   (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
   (e) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
   (f) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S6).
2. Does tree growth increase with longer growing seasons?
   (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
   (b) Longer thermal season increased growth for *B. papyrifera* (β = 0.44, UI_{90} [0.18, 0.7]), *B. populifolia* (β = 0.22, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* (β = 0.13, UI_{90} [0.01, 0.24]), but *A. incana* had a weak effect, but trended in the same direction (β = 0.07, UI_{90} [-0.04, 0.17]) (Figure 2a and e) (TableS1).
   (c) *A. incana* grew more with longer calendar seasons (β = 0.05, UI_{90} [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons (β = 0.18, UI_{90} [0.05, 0.32]).
   (d) *B. alleghaniensis* (β = 0.03, UI_{90} [-0.02, 0.09]), and *B. populifolia* (β = 0.05, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 2b and f) (Table S1).
   (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S4)
   (f) On a scaled axis, both predictors were similarly predictive (FigureS7).
3. Did a later start and end of season increase growth?
   (a) A longer calendar season increased growth for two species, but only one grew more with an earlier start of season (*A. incana*: β = -0.07, UI_{90} [-0.14, -0.01]) (Figure 2c and g) (Table S1).
   (b) *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction (β = 0.1, UI_{90} [0, 0.2]) (Figure 2c and g) (Table S1).
   (c) Three species grew more with a later end of season (*B. alleghaniensis*: β = 0.12, UI_{90} [0.05, 0.18], *B. populifolia*: β = 0.1, UI_{90} [0.02, 0.17]), with *B. papyrifera* (β = 0.21, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 2d and h) (Table S1).
   (d) The estimates were unchanged when using the full (SOS n = 239 and n = 230) or restricted datasets (n =227) (Figure S1a).
4. How did tree growth change across years, provenances and tree ids?

(a) We did not find any clear effect of year on growth (Figure S5) and there is no clear differences of growth looking across species (Figure S2 and Figure S3)

(b) Growth did not change across any provenances (Figure 3) (Table S2), and the variation across sites (σ = 0.18, UI_{90} [0.02, 0.5) and tree ids (σ = 0.17, UI_{90} [0.05, 0.27]) was similar

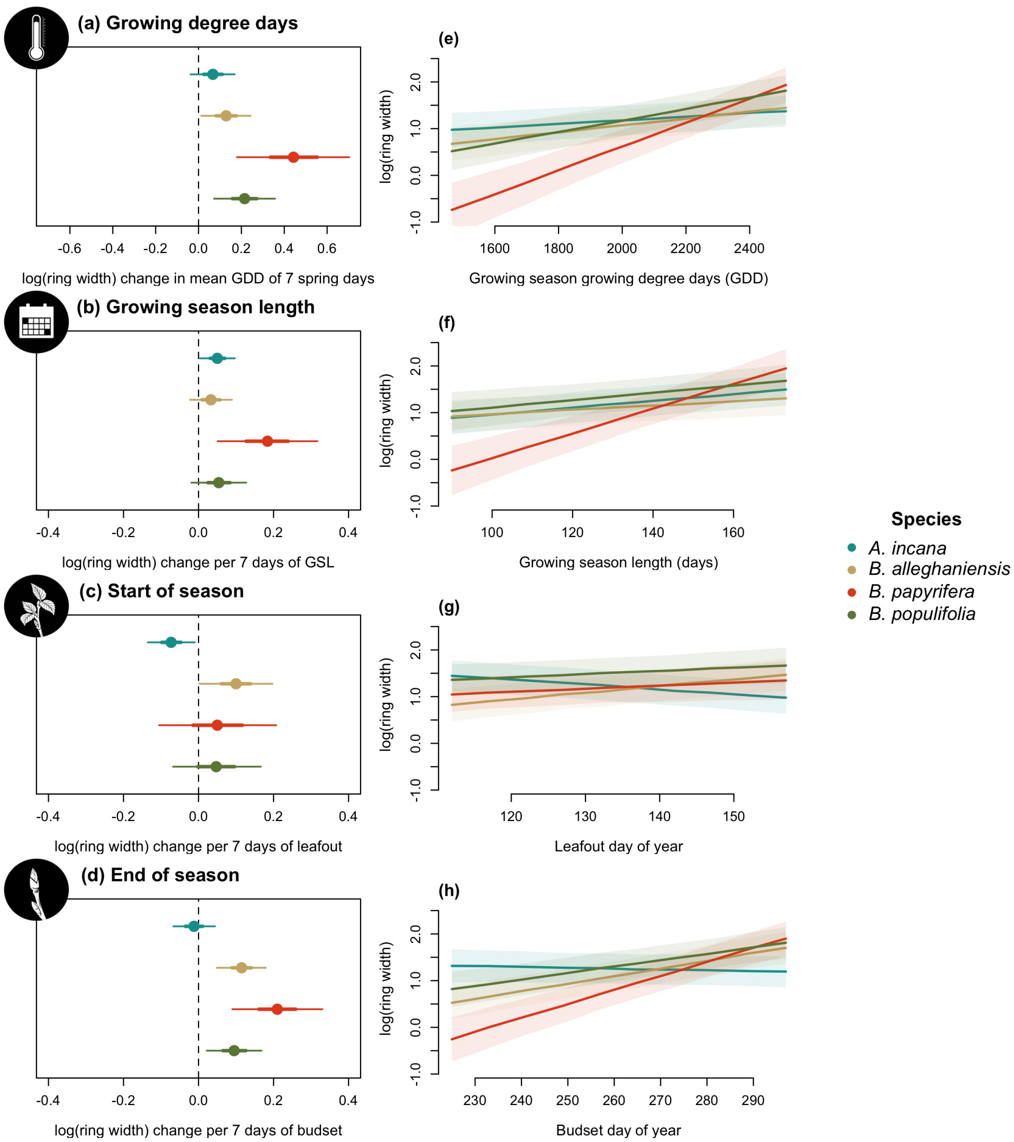

Figure 2: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (174.05) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

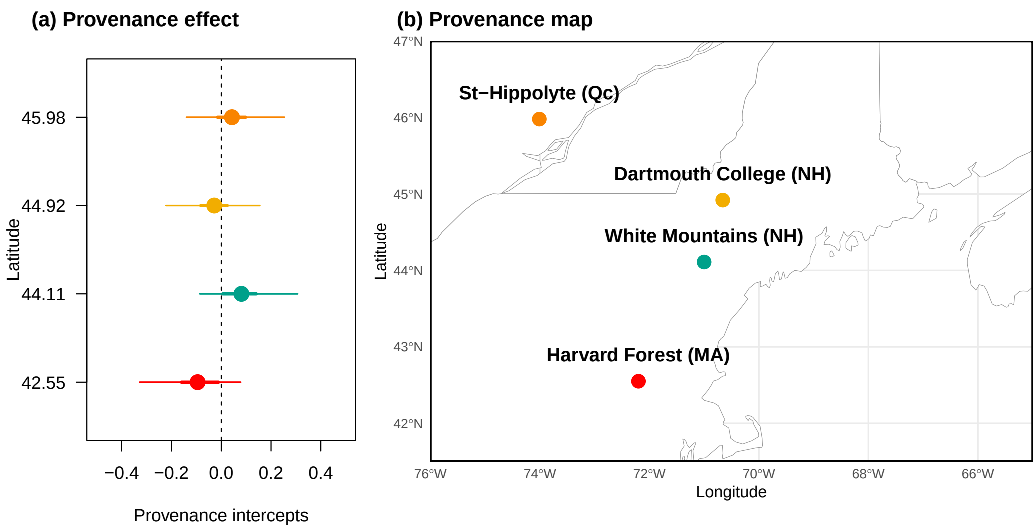

Figure 3: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Discussion

1. What we did, how we did it and what we found
   (a) We used 81 juvenile trees of four species across four provenances to understand if longer growing seasons using leaf phenology increased growth using tree rings
   (b) We found that longer growing seasons increased growth for all species, but varied depending on the metric
   (c) Thermal season was a better predictor for growth than the calendar season
2. Why thermal season better predicts growth than the calendar season?
   (a) Thermal weights growth days better than calendar season—trees don't grow at the same speed all year round
   (b) Cool and short spring days count for less in the thermal season compared to the calendar season
   (c) *A. incana* responding to GSL>GDD may be able to take advantage of cool and short days in spring more than other species
3. SOS and EOS to disentangle the calendar season effect on growth
   (a) *A. incana* grow more with GSL and earlier SOS, meaning that most growth happens in spring.
   (b) On the contrary, *B. papyrifera* driven by EOS rather than SOS
   (c) *B. alleghaniensis* Decrease in growth with earlier SOS and increase with later EOS counterinteract and lead to no response to GSL
4. Our trees grew more with longer seasons in absence of external constraints
   (a) When seasons are longer, but not drier, trees can fully take advantage of these extra days
   (b) Our study design controlled light competition—alleviating competition for light as a growth constraint
5. No effect of provenance
   (a) Our results suggest that local adaption on growth is weak under current conditions
   (b) Though our latitudinal gradient is narrow
6. Limitations
   (a) Low sample size and not a fully crossed study design
   (b) Didn't account for extreme temperatures (though no extended heat wave periods in our study)

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]	-74.03

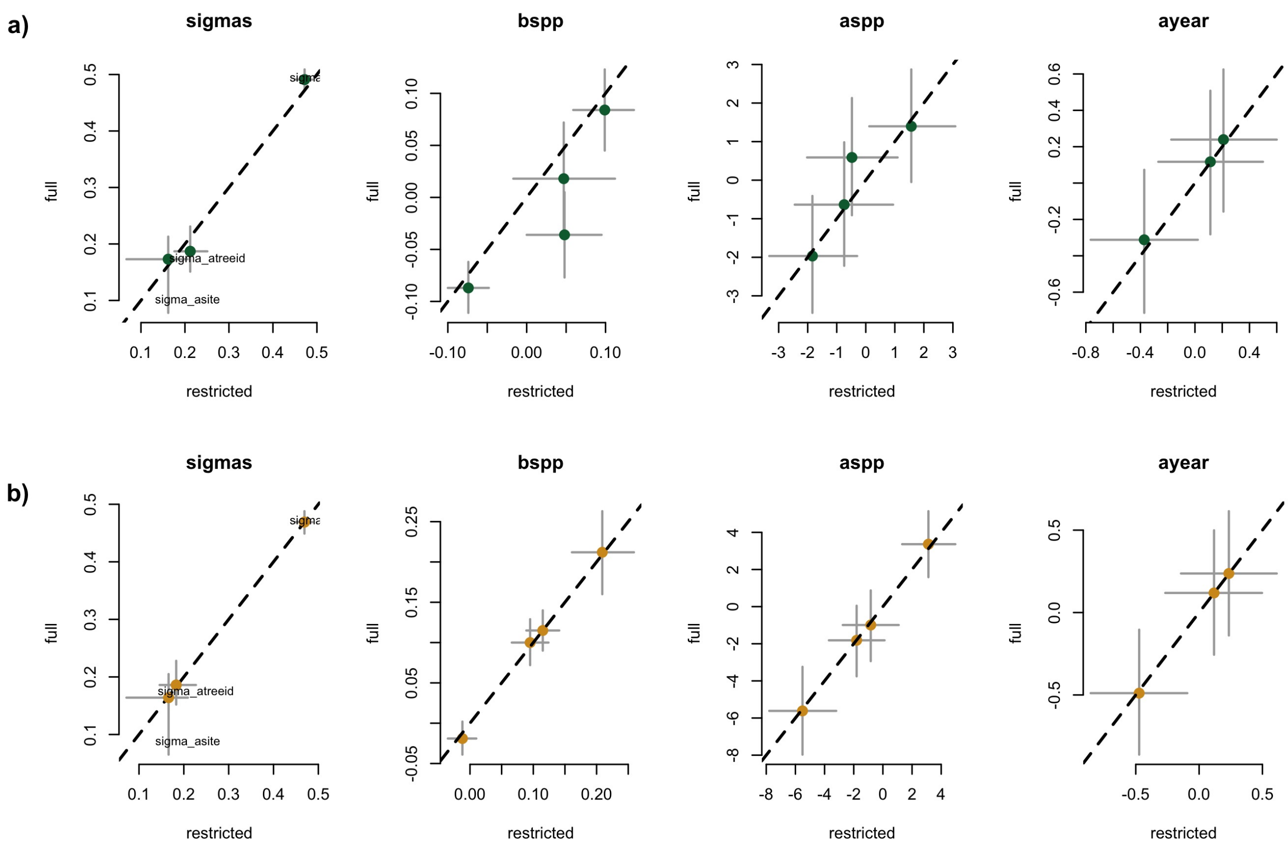

Figure S1: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

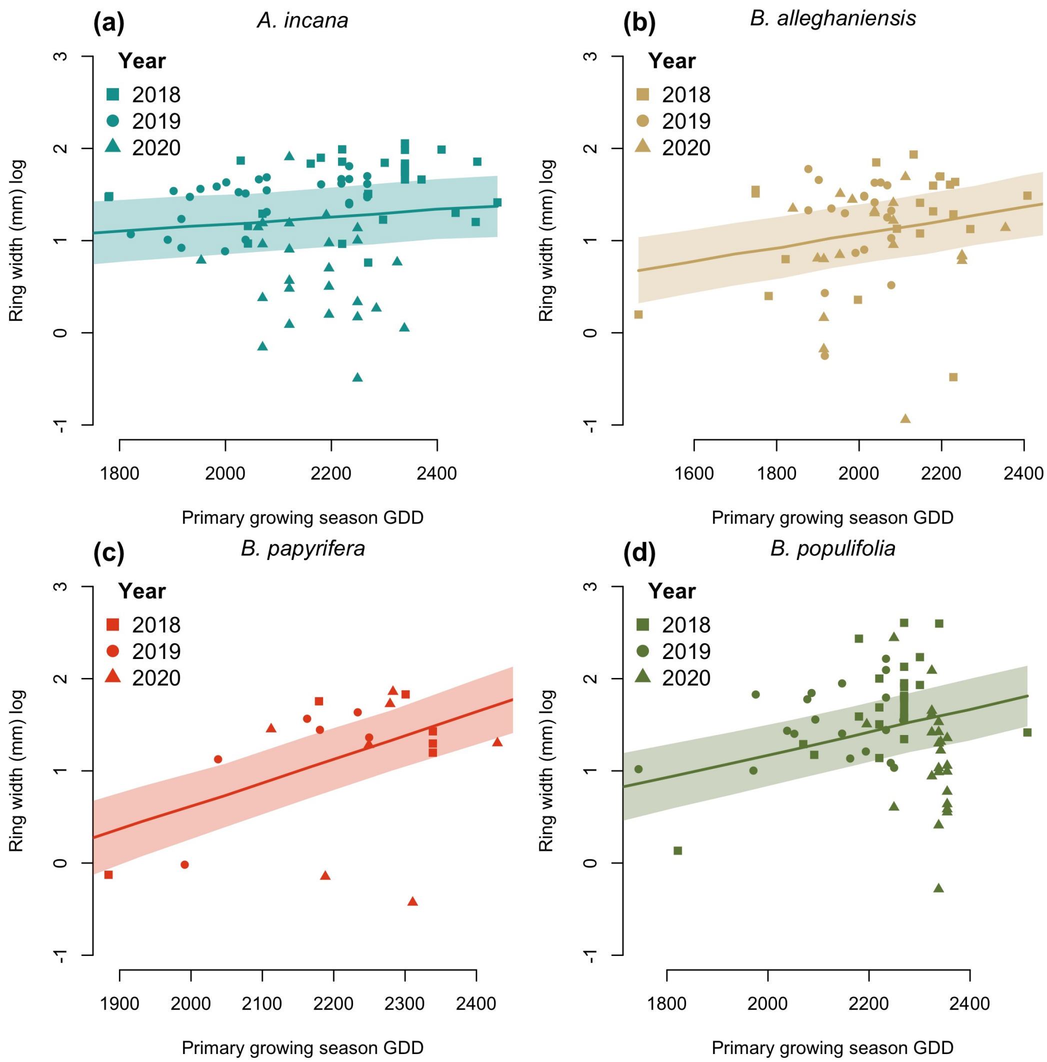

Figure S2: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

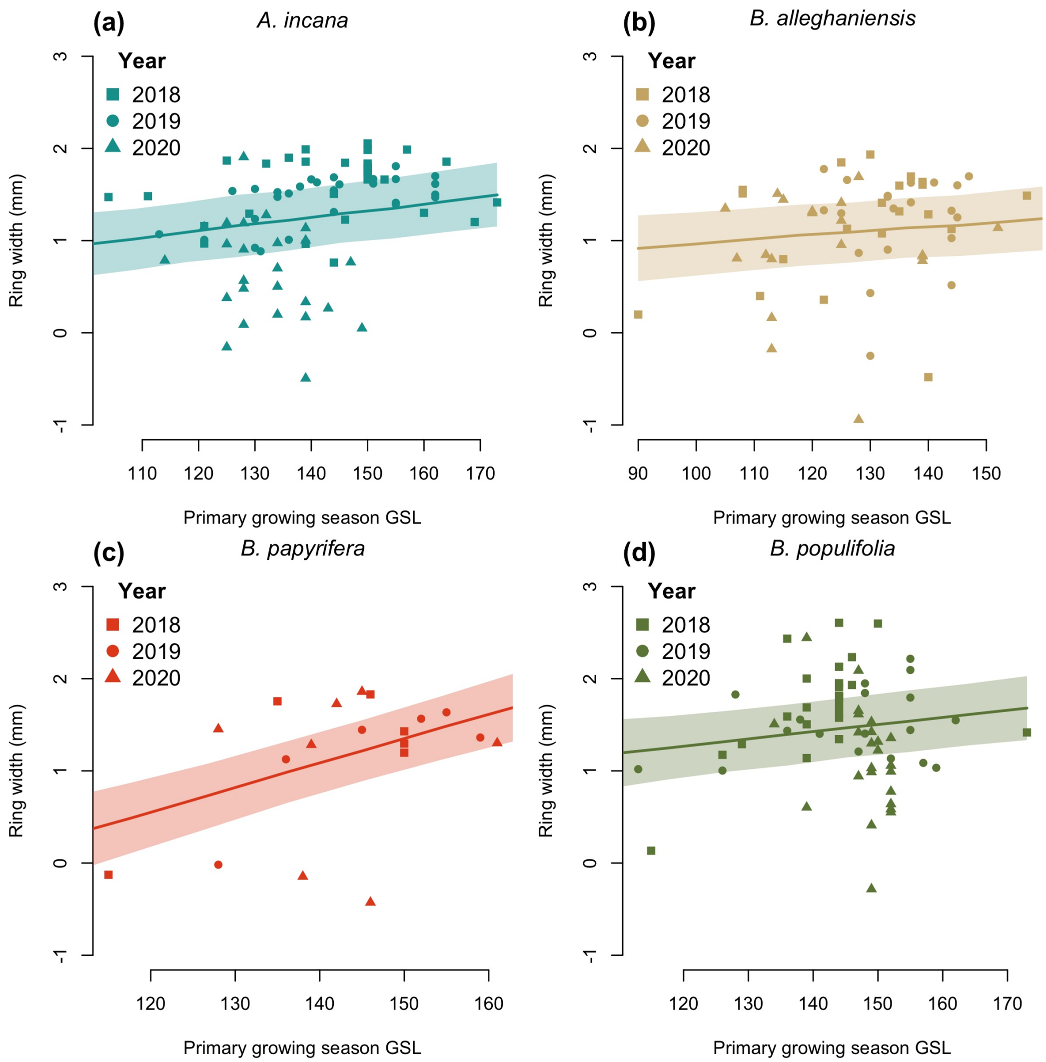

Figure S3: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

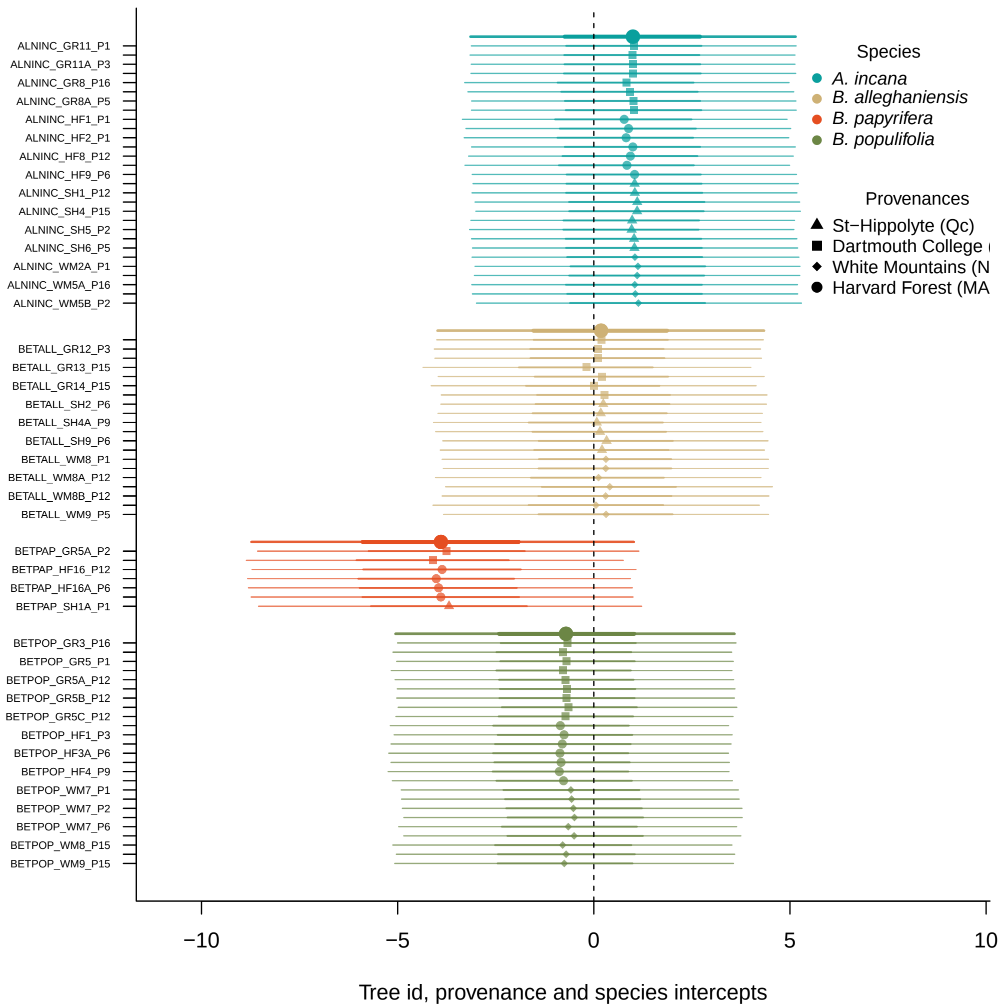

Figure S4: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

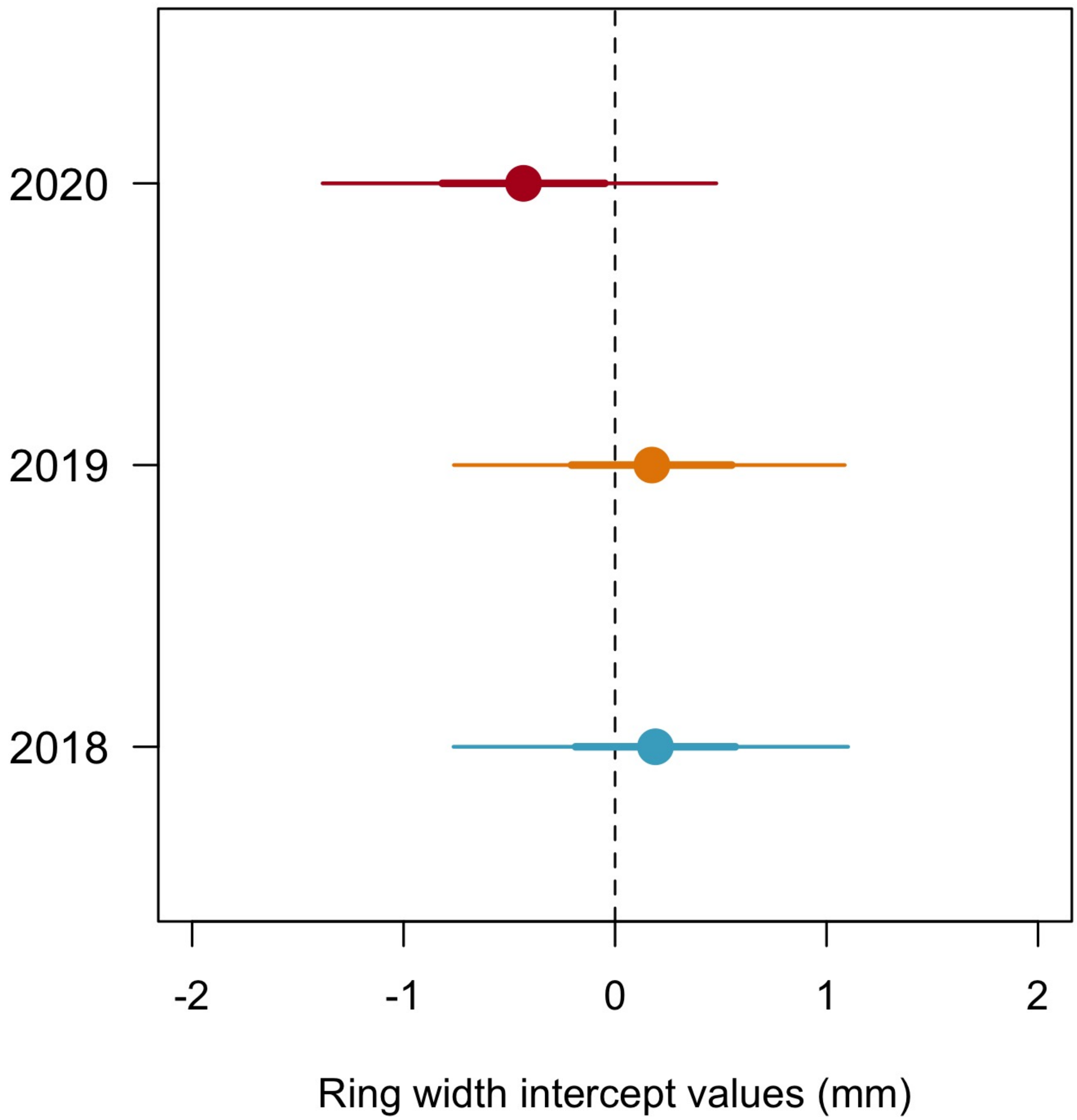

Figure S5: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

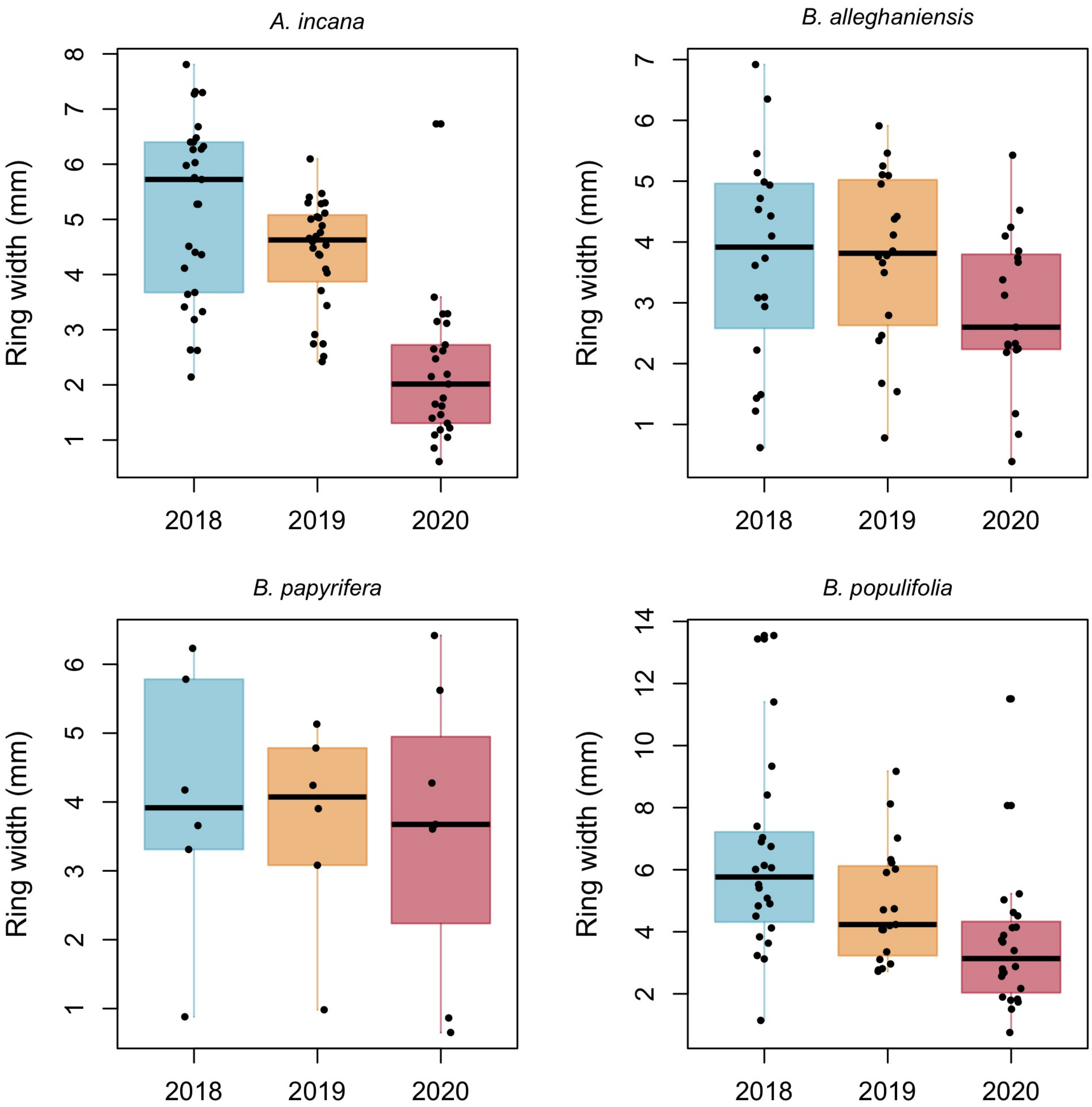

Figure S6: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

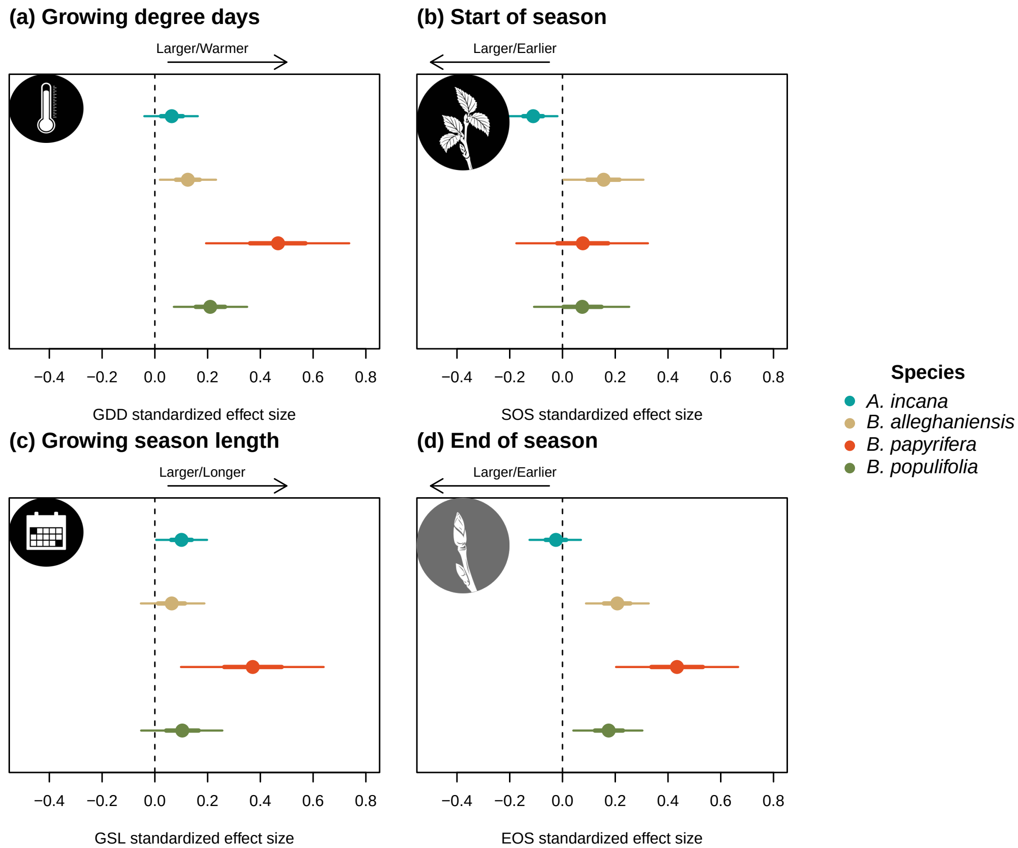

Figure S7: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters across predictors

Parameter	GDD	GSL	SOS	EOS
$\sigma_{\alpha_{tree}}$	0.17 [0.05, 0.27]	0.18 [0.05, 0.27]	0.21 [0.11, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{site}}$	0.18 [0.02, 0.50]	0.18 [0.02, 0.49]	0.16 [0.01, 0.45]	0.18 [0.03, 0.50]
σ_y	0.48 [0.43, 0.52]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{spp_{A.incana}}$	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{spp_{B.alleghaniensis}}$	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{spp_{B.papyrifera}}$	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
$\beta_{spp_{B.populifolia}}$	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]
$\alpha_{site_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]
$\alpha_{site_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]
$\alpha_{year_{2018}}$	0.19 [-0.76, 1.10]	0.20 [-0.77, 1.17]	0.20 [-0.72, 1.14]	0.23 [-0.71, 1.19]
$\alpha_{year_{2019}}$	0.17 [-0.76, 1.09]	0.07 [-0.89, 1.05]	0.10 [-0.83, 1.05]	0.12 [-0.83, 1.09]
$\alpha_{year_{2020}}$	-0.43 [-1.38, 0.48]	-0.38 [-1.35, 0.60]	-0.39 [-1.31, 0.56]	-0.47 [-1.42, 0.50]
$\alpha_{spp_{A.incana}}$	1.01 [-3.14, 5.18]	0.12 [-4.04, 4.32]	1.58 [-1.98, 5.48]	3.08 [-1.29, 7.36]
$\alpha_{spp_{B.alleghaniensis}}$	0.15 [-3.95, 4.36]	0.34 [-3.82, 4.49]	-1.82 [-5.51, 2.13]	-1.82 [-6.29, 2.64]
$\alpha_{spp_{B.papyrifera}}$	-3.82 [-8.73, 1.09]	-2.67 [-7.34, 1.89]	-0.76 [-4.84, 3.59]	-5.58 [-11.03, -0.04]
$\alpha_{spp_{B.populifolia}}$	-0.69 [-4.89, 3.51]	0.22 [-4.00, 4.45]	-0.42 [-4.24, 3.59]	-0.85 [-5.47, 3.82]

Table S4: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]

References

Aitken, S.N. & Bemmels, J.B. (2016) Time to get moving: assisted gene flow of forest trees. *Evolutionary Applications* **9**, 271–290.

Almagro, D., Martin-Benito, D., Rossi, S., Conde, M., Fernández-de-Uña, L. & Gea-Izquierdo, G. (2025) Long-Term Cambial Phenology Reveals Diverging Growth Responses of Two Tree Species in a Mixed Forest Under Climate Change. *Global Change Biology* **31**, e70503.

Cabon, A., Fernández-de-Uña, L., Gea-Izquierdo, G., Meinzer, F.C., Woodruff, D.R., Martínez-Vilalta, J. & De Cáceres, M. (2020) Water potential control of turgor-driven tracheid enlargement in Scots pine at its xeric distribution edge. *New Phytologist* **225**, 209–221.

Cabon, A., Kannenberg, S.A., Arain, A., Babst, F., Baldocchi, D., Belmecheri, S., Delpierre, N., Guerrieri, R., Maxwell, J.T., McKenzie, S., Meinzer, F.C., Moore, D.J.P., Pappas, C., Rocha, A.V., Szejner, P., Ueyama, M., Ulrich, D., Vincke, C., Voelker, S.L., Wei, J., Woodruff, D. & Anderegg, W.R.L. (2022) Cross-biome synthesis of source versus sink limits to tree growth. *Science* **376**, 758–761.

Change, I.P.O.C. (2014) Detection and Attribution of Climate Change: from Global to Regional. *Climate Change 2013 – The Physical Science Basis*, pp. 867–952, Cambridge University Press, 1 edn.

Chiang, F., Mazdiyasni, O. & AghaKouchak, A. (2021) Evidence of anthropogenic impacts on global drought frequency, duration, and intensity. *Nature Communications* **12**, 2754.

Chmielewski, F.M. & Rötzer, T. (2001) Response of tree phenology to climate change across Europe. *Agricultural and Forest Meteorology* **108**, 101–112.

Choat, B., Brodribb, T.J., Brodersen, C.R., Duursma, R.A., López, R. & Medlyn, B.E. (2018) Triggers of tree mortality under drought. *Nature* **558**, 531–539.

Chuine, I. (2010) Why does phenology drive species distribution? *Philosophical Transactions of the Royal Society B: Biological Sciences* **365**, 3149–3160.

Cleland, E., Chuine, I., Menzel, A., Mooney, H. & Schwartz, M. (2007) Shifting plant phenology in response to global change. *Trends in Ecology & Evolution* **22**, 357–365.

Cleland, E.E. & Wolkovich, E. (2024) Effects of Phenology on Plant Community Assembly and Structure. *Annual Review of Ecology, Evolution, and Systematics* **55**, 471–492.

Cooke, J.E.K., Eriksson, M.E. & Junttila, O. (2012) The dynamic nature of bud dormancy in trees: environmental control and molecular mechanisms. *Plant, Cell & Environment* **35**, 1707–1728.

Cuny, H.E., Fonti, P., Rathgeber, C.B., Von Arx, G., Peters, R.L. & Frank, D.C. (2019) Couplings in cell differentiation kinetics mitigate air temperature influence on conifer wood anatomy. *Plant, Cell & Environment* **42**, 1222–1232.

Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T. & Rathgeber, C.B.K. (2016) Temperate and boreal forest tree phenology: from organ-scale processes to terrestrial ecosystem models. *Annals of Forest Science* **73**, 5–25.

Dow, C., Kim, A.Y., D'Orangeville, L., Gonzalez-Akre, E.B., Helcoski, R., Herrmann, V., Harley, G.L., Maxwell, J.T., McGregor, I.R., McShea, W.J., McMahon, S.M., Pederson, N., Tepley, A.J. & Anderson-Teixeira, K.J. (2022) Warm springs alter timing but not total growth of temperate deciduous trees. *Nature* **608**, 552–557.

Dox, I., Skrøppa, T., Decoster, M., Prislan, P., Gascó, A., Gričar, J., Lange, H. & Campioli, M. (2022) Severe drought can delay autumn senescence of silver birch in the current year but advance it in the next year. *Agricultural and Forest Meteorology* **316**, 108879.

Duputié, A., Rutschmann, A., Ronce, O. & Chuine, I. (2015) Phenological plasticity will not help all species adapt to climate change. *Global Change Biology* **21**, 3062–3073.

Etzold, S., Sterck, F., Bose, A.K., Braun, S., Buchmann, N., Eugster, W., Gessler, A., Kahmen, A., Peters, R.L., Vitasse, Y., Walthert, L., Ziemińska, K. & Zweifel, R. (2022) Number of growth days and not length of the growth period determines radial stem growth of temperate trees. *Ecology Letters* **25**, 427–439.

Flynn, D.F.B. & Wolkovich, E.M. (2018) Temperature and photoperiod drive spring phenology across all species in a temperate forest community. *New Phytologist* **219**, 1353–1362.

Fu, Y.H., Piao, S., Op De Beeck, M., Cong, N., Zhao, H., Zhang, Y., Menzel, A. & Janssens, I.A. (2014) Recent spring phenology shifts in western Central Europe based on multiscale observations. *Global Ecology and Biogeography* **23**, 1255–1263.

Gallinat, A.S., Primack, R.B. & Wagner, D.L. (2015) Autumn, the neglected season in climate change research. *Trends in Ecology & Evolution* **30**, 169–176.

Gazol, A., Camarero, J.J., Vicente-Serrano, S.M., Sánchez-Salguero, R., Gutiérrez, E., De Luis, M., Sangüesa-Barreda, G., Novak, K., Rozas, V., Tíscar, P.A., Linares, J.C., Martín-Hernández, N., Martínez Del Castillo, E., Ribas, M., García-González, I., Silla, F., Camisón, A., Génova, M., Olano, J.M., Longares, L.A., Hevia, A., Tomás-Burguera, M. & Galván, J.D. (2018) Forest resilience to drought varies across biomes. *Global Change Biology* **24**, 2143–2158.

Gessler, A. & Zweifel, R. (2024) Beyond source and sink control – toward an integrated approach to understand the carbon balance in plants. *New Phytologist* **242**, 858–869.

Green, J.K. & Keenan, T.F. (2022) The limits of forest carbon sequestration. *Science* **376**, 692–693.

Housset, J.M., Nadeau, S., Isabel, N., Depardieu, C., Duchesne, I., Lenz, P. & Girardin, M.P. (2018) Tree rings provide a new class of phenotypes for genetic associations that foster insights into adaptation of conifers to climate change. *New Phytologist* **218**, 630–645.

Howe, G.T., Aitken, S.N., Neale, D.B., Jermstad, K.D., Wheeler, N.C. & Chen, T.H. (2003) From genotype to phenotype: unraveling the complexities of cold adaptation in forest trees. *Canadian Journal of Botany* **81**, 1247–1266.

Jeong, S. & Medvigy, D. (2014) Macroscale prediction of autumn leaf coloration throughout the continental United States. *Global Ecology and Biogeography* **23**, 1245–1254.

Kang, J., Yang, Z., Yu, B., Ma, Q., Jiang, S., Shishov, V.V., Zhou, P., Huang, J.G. & Ding, X. (2023) An earlier start of growing season can affect tree radial growth through regulating cumulative growth rate. *Agricultural and Forest Meteorology* **342**, 109738.

Keenan, T.F., Gray, J., Friedl, M.A., Toomey, M., Bohrer, G., Hollinger, D.Y., Munger, J.W., O'Keefe, J., Schmid, H.P., Wing, I.S., Yang, B. & Richardson, A.D. (2014) Net carbon uptake has increased through warming-induced changes in temperate forest phenology. *Nature Climate Change* **4**, 598–604.

Körner, C. & Basler, D. (2010) Phenology Under Global Warming. *Science* **327**, 1461–1462.

Körner, C., Möhl, P. & Hiltbrunner, E. (2023) Four ways to define the growing season. *Ecology Letters* **26**, 1277–1292.

Li, Y., Zhang, W., Schwalm, C.R., Gentine, P., Smith, W.K., Ciais, P., Kimball, J.S., Gazol, A., Kannenberg, S.A., Chen, A., Piao, S., Liu, H., Chen, D. & Wu, X. (2023) Widespread spring phenology effects on drought recovery of Northern Hemisphere ecosystems. *Nature Climate Change* **13**, 182–188.

Lieth, H., Jacobs, J., Lange, O.L., Olson, J.S. & Wieser, W. (eds.) (1974) *Phenology and Seasonality Modeling*, vol. 8 of *Ecological Studies*. Springer Berlin Heidelberg, Berlin, Heidelberg.

Manzanedo, R.D. & Pederson, N. (2019) Towards a More Ecological Dendroecology. *Tree-Ring Research* **75**, 152.

Marchand, L.J., Dox, I., Gričar, J., Prislan, P., Van Den Bulcke, J., Fonti, P. & Campioli, M. (2021) Timing of spring xylogenesis in temperate deciduous tree species relates to tree growth characteristics and previous autumn phenology. *Tree Physiology* **41**, 1161–1170.

Matula, R., Knířová, S., Vítámvás, J., Šrámek, M., Kníř, T., Ulbrichová, I., Svoboda, M. & Plichta, R. (2023) Shifts in intra-annual growth dynamics drive a decline in productivity of temperate trees in Central European forest under warmer climate. *Science of The Total Environment* **905**, 166906.

Menzel, A., Sparks, T.H., Estrella, N., Koch, E., Aasa, A., Ahas, R., Alm-Kübler, K., Bissolli, P., Braslavská, O., Briede, A., Chmielewski, F.M., Crepinsek, Z., Curnel, Y., Defila, C., Donnelly, A., Filella, Y., Jatczak, K., Mestre, A., Peñuelas, J., Pirinen, P., Scheifinger, H., Striz, M., Susnik, A., Van Vliet, A.J.H., Wielgolaski, F., Zach, S. & Zust, A. (2006) European phenological response to climate change matches the warming pattern. *Global Change Biology* **12**, 1969–1976.

Meyer, H.A. (1940) A Mathematical Expression for Height Curves. *Journal of Forestry* **38**, 415–420.

Michelot, A., Simard, S., Rathgeber, C., Dufrene, E. & Damesin, C. (2012) Comparing the intra-annual wood formation of three European species (Fagus sylvatica, Quercus petraea and Pinus sylvestris) as related to leaf phenology and non-structural carbohydrate dynamics. *Tree Physiology* **32**, 1033–1045.

Muller, B., Pantin, F., Génard, M., Turc, O., Freixes, S., Piques, M. & Gibon, Y. (2011) Water deficits uncouple growth from photosynthesis, increase C content, and modify the relationships between C and growth in sink organs. *Journal of Experimental Botany* **62**, 1715–1729.

Parmesan, C. & Yohe, G. (2003) A globally coherent fingerprint of climate change impacts across natural systems. *Nature* **421**, 37–42.

Peters, R.L., Steppe, K., Cuny, H.E., De Pauw, D.J., Frank, D.C., Schaub, M., Rathgeber, C.B., Cabon, A. & Fonti, P. (2021) Turgor – a limiting factor for radial growth in mature conifers along an elevational gradient. *New Phytologist* **229**, 213–229.

Peñuelas, J. & Filella, I. (2001) Responses to a Warming World. *Science* **294**, 793–795.

Piao, S., Liu, Q., Chen, A., Janssens, I.A., Fu, Y., Dai, J., Liu, L., Lian, X., Shen, M. & Zhu, X. (2019) Plant phenology and global climate change: Current progresses and challenges. *Global Change Biology* **25**, 1922–1940.

Plomion, C., Leprovost, G. & Stokes, A. (2001) Wood Formation in Trees. *Plant Physiology* **127**, 1513–1523.

Polgar, C.A. & Primack, R.B. (2011) Leaf-out phenology of temperate woody plants: from trees to ecosystems. *New Phytologist* **191**, 926–941.

Reinmann, A.B., Bowers, J.T., Kaur, P. & Kohler, C. (2023) Compensatory responses of leaf physiology reduce effects of spring frost defoliation on temperate forest tree carbon uptake. *Frontiers in Forests and Global Change* **6**, 988233.

Richardson, A.D., Keenan, T.F., Migliavacca, M., Ryu, Y., Sonnentag, O. & Toomey, M. (2013) Climate change, phenology, and phenological control of vegetation feedbacks to the climate system. *Agricultural and Forest Meteorology* **169**, 156–173.

Saunders, M.R. & Wagner, R.G. (2008) Height-diameter models with random coefficients and site variables for tree species of Central Maine. *Annals of Forest Science* **65**, 203–203.

Silvestro, R., Deslauriers, A., Prislan, P., Rademacher, T., Rezaie, N., Richardson, A.D., Vitasse, Y. & Rossi, S. (2025) From Roots to Leaves: Tree Growth Phenology in Forest Ecosystems. *Current Forestry Reports* **11**, 12.

Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S. & Rossi, S. (2023) A longer wood growing season does not lead to higher carbon sequestration. *Scientific Reports* **13**, 4059.

Soolanayakanahally, R.Y., Guy, R.D., Silim, S.N. & Song, M. (2013) Timing of photoperiodic competency causes phenological mismatch in balsam poplar (*Populus balsamifera* L.). *Plant, Cell & Environment* **36**, 116–127.

Stridbeck, P., Björklund, J., Fuentes, M., Gunnarson, B.E., Jönsson, A.M., Linderholm, H.W., Ljungqvist, F.C., Olsson, C., Rayner, D., Rocha, E., Zhang, P. & Seftigen, K. (2022) Partly decoupled tree-ring width and leaf phenology response to 20th century temperature change in Sweden. *Dendrochronologia* **75**, 125993.

Suzuki, M., Yoda, K. & Suzuki, H. (1996) Phenological Comparison of the Onset of Vessel Formation Between Ring-Porous and Diffuse-Porous Deciduous Trees in a Japanese Temperate Forest. *IAWA Journal* **17**, 431–444.

Swidrak, I., Schuster, R. & Oberhuber, W. (2013) Comparing growth phenology of co-occurring deciduous and evergreen conifers exposed to drought. *Flora: Morphology, Distribution, Functional Ecology of Plants* **208**, 609–617.

Trenberth, K.E., Dai, A., Van Der Schrier, G., Jones, P.D., Barichivich, J., Briffa, K.R. & Sheffield, J. (2014) Global warming and changes in drought. *Nature Climate Change* **4**, 17–22.

Tyree, M.T. & Zimmermann, M.H. (2002) *Xylem Structure and the Ascent of Sap.* Springer Series in Wood Science, Springer Berlin Heidelberg, Berlin, Heidelberg.

Vitasse, Y., eLenz, A. & eKoerner, C. (2014) The interaction between freezing tolerance and phenology in temperate deciduous trees. *Frontiers in Plant Science* **5**.

Wang, W., Qin, L., Zhang, T., Yang, F., Cabon, A., Wang, Z., Zhou, P., Zhang, Y., Fonti, P. & Huang, J. (2025) Seasonal climate variations drive decoupling between the duration and amount of xylem growth along a hydrothermal gradient in the southern Altai Mountains. *Journal of Ecology* **113**, 1793–1806.

Way, D.A. & Montgomery, R.A. (2015) Photoperiod constraints on tree phenology, performance and migration in a warming world. *Plant, Cell & Environment* **38**, 1725–1736, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/pce.12431.

Wolfe, D.W., Schwartz, M.D., Lakso, A.N., Otsuki, Y., Pool, R.M. & Shaulis, N.J. (2005) Climate change and shifts in spring phenology of three horticultural woody perennials in northeastern USA. *International Journal of Biometeorology* **49**, 303–309.

Wolkovich, E.M., Ettinger, A.K., Chin, A.R., Chamberlain, C.J., Baumgarten, F., Pradhan, K., Manzanedo, R.D. & Hille Ris Lambers, J. (2025) Why longer seasons with climate change may not increase tree growth. *Nature Climate Change* **15**, 1283–1292.

Woolway, R.I., Sharma, S., Weyhenmeyer, G.A., Debolskiy, A., Golub, M., Mercado-Bettín, D., Perroud, M., Stepanenko, V., Tan, Z., Grant, L., Ladwig, R., Mesman, J., Moore, T.N., Shatwell, T., Vanderkelen, I., Austin, J.A., DeGasperi, C.L., Dokulil, M., La Fuente, S., Mackay, E.B., Schladow, S.G., Watanabe, S., Marcé, R., Pierson, D.C., Thiery, W. & Jennings, E. (2021) Phenological shifts in lake stratification under climate change. *Nature Communications* **12**, 2318.

Zeng, Z.A. & Wolkovich, E.M. (2024) Weak evidence of provenance effects in spring phenology across Europe and North America. *New Phytologist* **242**, 1957–1964.